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Local LLM-Driven Balance Mediation in Competitive Gaming

1. Introduction

Each balance patch for a competitive game typically provokes debate within its community. Players express their opinions using emotional language, community-specific slang, and multiple languages, requiring designers to interpret specific requests regarding characters and moves, as well as whether changes should strengthen or weaken them. The project plan dated September 20, 2026, proposed evaluating whether small, locally hosted large language models (LLMs) could serve as mediators between players and designers. The plan established four objectives: to digest community feedback, filter out emotional bias, map domain-specific terminology, and calculate a consensus on the adjustments requested by players. Technically, this approach constitutes aspect-based opinion mining [9, 10], where the aspects are characters and moves, with the additional requirement that each result is traceable to its source.

The resulting product is AI Mediator, an installable command-line tool (AI Mediator) designed to process Street Fighter 6 feedback regarding the 08.03.2026 Update, the balance patch released on August 3, 2026 [1]. As specified in the plan, English comments are processed by Llama 3.1 8B [3] and Chinese comments by Qwen 2.5 7B [2], both hosted via Ollama [5]. The tool ingests a file of collected comments and generates a “Community Consensus Report” that, for each character, enumerates the supported buff and nerf requests, each linked to the precise sentence from which it was derived. A seven-stage pipeline processes the feedback (Section 2.4), including an evidence-quality stage that distinguishes venting from substantive points, and a bilingual glossary that maps terminology across 849 game terms and 4,332 community aliases.

Three parts of the plan changed. First, the sources: comments were collected through the official Steam review and YouTube APIs instead of from Reddit and Bilibili. Reddit only allows spiders to extract comments from half a year ago, which means the data is outdated. On the other hand, Bilibili doesn’t allow unauthorized data extraction. Second, the consensus: the plan promised “an objective mathematical consensus on required mechanical adjustments”, but experiments showed that few comments contain numbers and that small models readily invent them (Section 3). The tool therefore counts the distinct comments behind each (character, direction) position, keeps opposing views side by side, and reports a numerical change only when a comment explicitly requests one. The counts describe the collected sample, not the entire player base. Third, the models: the local models proved too slow and unreliable on the available hardware, and the final evaluation used a hosted model (deepseek-v4-flash) through an API (Section 4). The scope remained limited to one game, one patch, two languages, and a six-week budget at five hours per week.

2. Structure of the Program and Technical Details

2.1 Overview

The program is a Python package (Python 3.9 or later, NumPy as the only dependency) of about thirty modules in layers: the command-line interface, services, the pipeline, infrastructure (model calls, caching, run storage, retrieval), and domain rules (evidence, glossary, response schemas). A test checks that each module imports only from its own layer or the layers below. The commands are init and doctor (configuration and a check of the model service), analyze and status (a resumable run and its progress), review export, review import and report (a human or agent reviewer decides on every proposed claim; automatic or reviewed reports are exported), evaluate (scoring of quality annotations), and demo (an offline run on 38 synthetic bilingual comments that needs no model and takes about one second). The package has 333 offline tests. The pipeline runs in this order:

```
comments -> (1) language routing -> (2) admission -> (3) evidence quality ->
   (4) claim extraction -> (5) source binding -> (6) semantic audit -> (7)
                            aggregation -> report
```

2.2 Data Collection

The collectors call official APIs with bounded request counts and fixed spacing: the Steam review API [16] and the YouTube Data API [17]; Bluesky refused access (HTTP 403). Raw records are never modified; a coarse keyword screen, duplicate removal, and recorded language corrections produce the screened sets. The expanded set (October 15) holds 321 comments: 207 English, 113 Chinese, and 1 Portuguese, which is recorded and excluded. The balanced set (October 27) holds 600 comments: 300 English and 300 simplified Chinese, all Steam reviews, selected from 6,376 fetched records. Its English reviews date from December 2025 to October 2026 and its Chinese ones from August to October 2026, so not every comment refers to the target patch.

2.3 Bilingual Glossary

The glossary has 849 entries and 4,332 aliases covering all 31 characters of the target patch and the system, mechanic, input, and move names; 157 alias records are marked ambiguous. Move identities come from Capcom’s official pages and Chinese move names from community references. The glossary is embedded with Qwen3-Embedding-8B [4] into a vector index that the program searches, in the manner of retrieval-augmented generation [8]. An alias match is only a candidate, never proof of identity: nationality nicknames once produced false matches for one character, and an identity policy now blocks them.

2.4 Pipeline Stages

Stage 1: Routing. English comments and English instructions go only to Llama; Chinese comments and equivalent Chinese instructions go only to Qwen. Comments are never translated or rewritten.

Stage 2: Admission. Each comment is embedded and compared with ten positive and ten negative descriptions of combat-balance topics; the mean similarity of the three nearest positive topics must exceed that of the three nearest negative ones. The routed model then judges whether the comment is about those topics. Only comments that pass both checks reach extraction.

Stage 3: Evidence Quality (Emotion Filter). This optional stage runs in one of three modes: off, shadow (measure only), or enforce. The model labels comment segments by scope (character

balance, system balance, or unrelated), content kind (pure emotion, general opinion, or concrete feedback), and emotional tone. The program, not the model, then computes:

```
emotion_ratio = W(segments labeled pure emotion) / W(whole comment)
```

where W counts Unicode letters, numbers, marks, and symbols [18], excluding spaces and punctuation; a ratio of 0.70 or more raises a warning. Each segment is routed to exclude (pure venting), opinion only, mechanism evidence, system evidence, or review. The central rule is that emotion never lowers the weight of a concrete point: an angry comment that names a real problem keeps the problem, and only the venting is set aside. In enforce mode, only mechanism evidence can support a claim.

Stage 4: Extraction. The model answers in JSON constrained by a schema [14]: a list of claims per comment, each with its supporting evidence (chosen first), character, target, direction (buff, nerf, keep, or unclear), recommendation, and any numerical parameters. A parameter is allowed only when the comment asks for a numerical change: “Ryu is too strong” supports a general nerf request, not a damage figure.

Stage 5: Source Binding. The program rebuilds every quotation from its evidence ID and checks it against the original comment's Unicode offsets. It rejects numbers absent from the source, characters the comment does not name, and directions that contradict the stated action.

Stage 6: Semantic Audit. A second request, in a fresh context, asks whether the full comment supports the whole claim. Unsupported or uncertain claims are quarantined with their reason.

Stage 7: Aggregation. Plain code builds the report: claims grouped by character and direction, with their evidence, the number of distinct comments behind each group, and the opposing views. Positions are not merged into numbers.

Every model request and response is saved and content-addressed, so an interrupted run resumes without repeating work, and each run records the hashes of its data, prompts, glossary, code and model weights. Every admitted comment receives exactly one coverage record (completed, rejected, pending or failed), so no comment disappears silently.

3. Experiment Data

3.1 Evaluation Sets and Measures

We used two labeled sets: balance-silver-v1 (100 comments, 50 English and 50 Chinese, split 70 development and 30 held out) for relevance, and extraction-precision-v1 (227 development and 74 test comments) for labeled (character, direction) positions. A set of 38 synthetic comments tests the emotion routes, and samples of 30 and 60 real comments measure speed. An AI agent wrote the labels from the source text (“silver” labels); the held-out and test splits were run once at the end. Precision is the share of accepted positions that match a label; recall is the share of labeled positions found. Intervals are Wilson 95% intervals [15], and difference intervals are Newcombe intervals [20]. The project repository contains the full record of every run [19].

3.2 Experiment 1: Expanded Set, Local Models (October 22)

Table 1. Full run on 321 comments

Measure	Value
Admitted (retrieval and relevance agree)	197 (112 English, 85 Chinese)
Complete claim proposals	122 (2 comments failed generation)
Failed source binding	4 (3 copied an “8 to 6 frames” example from the prompt)
Kept by the automatic audit	28 of 118
Kept after a full source review	47 claims from 38 comments, 15 characters
Numerical entailment controls	3 of 4

The source review rejected 8 of the 28 claims the audit had accepted. Typical errors: cosmetic praise read as balance feedback, a quoted old patch read as a new request, and attacker and victim swapped. All 47 retained claims were qualitative. In a control, Qwen accepted “隆的波动拳伤害为600，肯的波动拳伤害为500” (“Ryu’s Hadoken does 600 damage, Ken’s does 500”) as support for changing Ryu’s damage from 600 to 500. These are forms of hallucination, output that the input does not support [11].

3.3 Experiment 2: Balanced Set, Local Models (October 27)

Table 2. Full run on 600 comments (emotion stage off)

Measure	Value
Admitted	525
Claims generated	404
Quarantined by source binding	17
Rejected by the semantic audit	200 of 387 (51.7%, 46.7-56.6)
Accepted	187 (87 English, 100 Chinese)
Comments with failed extraction	2
Synthetic controls	numerical 6 of 6, semantic 8 of 8

Rejection was highest for Chinese claims with an unclear direction (53 of 62) and for comments yielding three or more claims (64%). A sample of rejected claims showed mostly real errors, such as passing mentions in long pros-and-cons reviews turned into nerf requests. A retry of one long

comment produced 10 claims that the audit accepted and a source review rejected, all attributing World Tour and Drive Impact complaints to Ryu: a model checking its own output tends to miss its own errors [12]. A separate audit of the expanded set found relevance admission precision of 98.1% (264 of 269) and recall of 88.3% (264 of 299), but also found the emotion stage deleting concrete feedback along with the venting around it, and reading sarcasm (“another year of Ryu buffs”) as a mechanism.

3.4 Final Acceptance Runs (November 5, Hosted Model)

Table 3. Development results and the final held-out runs

Measure	Development	Held out (run once)
Relevance agreement	64/69 (92.8%)	27/29 (93.1%)
Precision of accepted claims	58/77 (75.3%)	19/32 (59.4%, 42.3-74.5)
Recall of labeled positions	58/123 (47.2%)	19/71 (26.8%, 17.9-38.1)
Claims rejected by the audit	33.6%	50.7%
English precision; recall	30/33; 30/46	10/12; 10/17
Chinese precision; recall	28/44; 28/77	9/20; 9/54

The drop on the test split comes mostly from Chinese comments, which hold 76% of its labeled positions. We tuned nothing after the test result. One run cannot separate a harder split from overfitting. Still, we reverted the prompt changes made against the development data (Section 4.4), and the pre-upgrade reference scores were about as far above the test split as the final code. In enforce mode, only 1 of 30 benchmark comments contained a concrete mechanism segment, and it yielded no claim; the labeled requests sat in comments routed as general opinions (“buff X”), which the report lists in its opinion summary. The filter works as designed, but in its strict form it leaves the formal claim list almost empty for typical Steam reviews.

4. Failure of the Local LLMs and the Move to a Hosted API

4.1 Local Setup

4.2 How the Local Model Failed

All local runs used my laptop: an AMD Radeon RX 6700S GPU with 8 GiB of memory, 24 GiB of system memory, and Ollama 0.34.2, with temperature 0, a fixed seed, and an 8,192-token context chosen so that the chat models fit in GPU memory. The 8B embedding model did not fit: only 27 to 29 of its 37 layers ran on the GPU, and it traded GPU memory with the chat models during a run.

4.2 How the local model failed

- **Speed:** An extraction request took about 20 seconds, and extraction alone took 40 minutes of the 600-comment run. With the emotion stage on, a 60-comment benchmark took 52 minutes instead of 7, because Llama wrote long answers and each quality request took about 40 seconds.
- **Format reliability:** Only 75.9% (41 of 54) of first-quality annotations passed validation, and 13 of the 60 comments had no usable annotation even after a retry. Seven extraction requests of the 600-comment run were stopped because the model repeated the same claim over and over [13], and one long comment failed every recovery attempt.
- **Accuracy:** The audit rejected half of all generated claims (51.7%). On the development set, the local models reached 55.6% precision and 28.8% recall, and the emotion routes matched the expected routes on only 16 of 34 synthetic comments (6 of 18 in Chinese).
- **Stability:** An early GPU backend failure required a CPU fallback option; the service's 10-second connection timeout had to be raised to 120 seconds, and in the early hours of November 5 the laptop shut down from overload in the middle of the local evaluation runs. The same shutdown happened at least 5 times from October 27-29. The evaluation plan needed dozens of runs of close to an hour each, so a local-only evaluation was not feasible.

4.3 The API Bridge

Rather than change the program, a small bridge was written in standard-library Python to answer the four Ollama endpoints the program uses and forward each request to OpenRouter [6]. Both chat model names map to deepseek/deepseek-v4-flash [7] on one pinned provider, and the embedding model maps to the hosted build of Qwen3-Embedding-8B. Messages pass through unchanged. The bridge reports its own version and model digests, so every run records that it used the API; it logs counts, timings, and cost but never content, and it refuses requests once spending reaches a 3.00 USD limit. The package and its tests did not change, and the reference runs were repeated through the API so every comparison used the same model on both sides.

Table 4. The same inputs and code, local models against the hosted model

Measure	Local (Llama, Qwen)	Hosted API
Wall time of the five reference runs	63.1 min	19.1 min
Emotion-stage time per comment	46.9 s	18.3 s
Extraction precision (114 comments)	68.2%	76.2%
Extraction recall	32.5%	48.8% (+16.3 points, 95% CI +4.0 to +27.9)
Claims rejected by the audit	54.8%	38.8%
Relevance agreement (69 comments)	59	63
Quality route agreement (synthetic)	8/18	11/16

4.4 Results of the Upgrade Through the API

With the hosted model, the November upgrade cut the extraction prompt from 4,064 to 2,788 tokens (31% less) and the quality prompt from 3,469 to 1,926 tokens (44% less). Emotion-stage time fell from 18.3 to 9.7 seconds per comment, and a 30-comment run fell from 104 to 44 seconds with two parallel requests, with the same accepted claims on all 30 comments. All upgrade evaluation runs together cost about 0.37 USD. Two changes that looked promising were measured, failed, and reverted: letting the program cut comments into clauses for the quality labels (more failed comments, first-attempt validity 78.3% against 87.5%), and adding worked examples and a subject list to the extraction prompt (precision 73.2% against 75.3%, recall 42.3% against 47.2%).

One hosted model now answers both languages, so the final figures describe that model, not Llama or Qwen. The package still routes to the local models by default, and its offline functions need no model. The project's starting premise, that 7B-8B models on a consumer laptop can run this pipeline, did not hold.

5. Conclusion

The project delivered the planned tool: an installable command-line program that admits on-topic bilingual feedback, separates venting from concrete points, extracts buff and nerf claims, checks every claim against the exact source text, and writes a report that counts the comments behind each position. Every entry in the report can be traced to a sentence a player wrote, and no comment is dropped without a recorded reason.

The results also mark the limits of the approach. On the final test split, 59.4% of accepted claims were right, and the model found 26.8% of the labeled requests, with Chinese comments well behind English ones. The plan's "objective mathematical consensus" became comment counts in a sample, because players rarely give numbers and the models invent them when pushed. The emotion filter is measurable and transparent, but in its strict mode most real comments turn out to be general opinions rather than concrete evidence. The labels are AI-written silver labels, and the audit uses the same model that wrote the claims, so the figures are bounds, not proof.

The clearest lesson concerns deployment. Local 7B-8B models on an 8 GiB laptop GPU were not adequate for a multi-stage pipeline with strict output; a hosted model was about three times faster and 16 points better in recall on the same inputs and code. Deterministic program checks mattered more than prompt wording: they caught copied example numbers, unsupported quotations, and impossible values, while the prompt changes that looked good on development data did not survive measurement. Future work: human double annotation of the evaluation sets, an independent second audit model, better Chinese extraction, the Reddit and Bilibili sources of the original plan, verified move data and patch dates, and a new attempt at local deployment on stronger hardware.

References

[1] Capcom. Street Fighter 6 battle changes: 08.03.2026 Update. https://www.streetfighter.com/6/buckler/battle_change/20260803

[2] Qwen Team (A. Yang et al.). Qwen2.5 Technical Report. arXiv:2412.15115, 2024.

[3] A. Grattafiori et al. (Llama Team, AI @ Meta). The Llama 3 Herd of Models. arXiv:2407.21783, 2024.

[4] Y. Zhang et al. Qwen3 Embedding: Advancing Text Embedding and Reranking Through Foundation Models. arXiv:2506.05176, 2025.

[5] Ollama. Ollama and its REST API documentation. https://github.com/ollama/ollama

[6] OpenRouter. API documentation. https://openrouter.ai/docs

[7] DeepSeek. DeepSeek V4 Flash, model listing on OpenRouter. https://openrouter.ai/deepseek/deepseek-v4-flash (accessed November 5, 2026).

[8] P. Lewis et al. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. NeurIPS 2020. arXiv:2005.11401.

[9] M. Pontiki et al. SemEval-2014 Task 4: Aspect Based Sentiment Analysis. Proceedings of SemEval 2014, pp. 27-35, 2014.

[10] B. Liu. Sentiment Analysis and Opinion Mining. Morgan & Claypool, 2012.

[11] Z. Ji et al. Survey of Hallucination in Natural Language Generation. ACM Computing Surveys 55(12), article 248, 2023.

[12] L. Zheng et al. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. NeurIPS 2023 Datasets and Benchmarks Track. arXiv:2306.05685.

[13] A. Holtzman et al. The Curious Case of Neural Text Degeneration. ICLR 2020. arXiv:1904.09751.

[14] B. T. Willard and R. Louf. Efficient Guided Generation for Large Language Models. arXiv:2307.09702, 2023.

[15] E. B. Wilson. Probable Inference, the Law of Succession, and Statistical Inference. Journal of the American Statistical Association 22(158), pp. 209-212, 1927.

[16] Valve. Steamworks documentation: User Reviews - Get List. https://partner.steamgames.com/doc/store/getreviews

[17] Google. YouTube Data API v3: CommentThreads. https://developers.google.com/youtube/v3/docs/commentThreads

[18] The Unicode Consortium. Unicode Standard Annex #44: Unicode Character Database. https://www.unicode.org/reports/tr44/

[19] W. Yang. AI Mediator project repository: README.md, docs/design/balance-analysis.md, docs/records/20261027-balanced-validation.md, docs/records/20261104-pipeline-upgrade.md, docs/records/20261105-pipeline-upgrade-validation.md. 2026.

[20] R. G. Newcombe. Interval Estimation for the Difference Between Independent Proportions: Comparison of Eleven Methods. Statistics in Medicine 17(8), pp. 873-890, 1998.